# A Comparative Analysis of Capacitor-Less LDO Topologies in 130nm CMOS Technology

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***Abstract*—This paper presents the design and comparison of three capacitor-less (CL) PMOS low dropout (LDO) regulators implemented in a 130nm CMOS technology. Each LDO topology was designed using a different OTA (5T OTA, Current Mirror OTA, Telescopic OTA). All three LDO configurations drive a 50 pF load, deliver a load current of $500\,\mu\mathrm{A}$, support programmable output voltages ($2.3\,\mathrm{V}$, $2.1\,\mathrm{V}$ and $1.6\,\mathrm{V}$), and have a consumption strictly below $100\,\mu\mathrm{A}$. Simulation results demonstrate that the 5T OTA provides the most balanced final results, whereas the Telescopic OTA maximizes loop gain at the sever expense of the Error Amplifier's output voltage swing. Furthermore, the Current Mirror OTA is shown to be incapable of satisfying the target specifications while keeping all the transistors in saturation.**

## I. Introduction and Theoretical Framework

Low dropout (LDO) voltage regulators are one of the most essential building blocks in power-management systems. Specially with today's increasing demand for system-on-chip solutions, many mixed-signal systems utilize LDO regulators to deliver local supply voltages with reduced noise.

This work will focus on designing three individual LDO architectures. Each LDO topology must meet the proposed specifications that are presented in Table I. The general LDO schematic is also shown in Figure 1.

TABLE I
Technical LDO Target Specifications

| Parameter | Proposed Value |
|---|---|
| Power Supply Voltage | $3\,\mathrm{V}$ |
| Bias Current $I_{bias}$ | $5\,\mu\mathrm{A}$ |
| Load Current $I_L$ | $500\,\mu\mathrm{A}$ |
| Load Capacitance $C_L$ | $50\,\mathrm{pF}$ |
| Reference Voltage $V_{REF}$ | $1.2\,\mathrm{V} \pm 1\,\%$ |
| Output Voltage $V_{OUT}$ | $2.3\,\mathrm{V}$ |
| Trim Voltages | $2.1\,\mathrm{V}$, $1.6\,\mathrm{V}$ |
| Compensation Capacitor $C_c$ | $<7\,\mathrm{pF}$ |
| Phase Margin @ No-Load | $>60°$ |
| Phase Margin @ Full-Load | $>70°$ |
| PSRR @ 1 Hz | $>65\,\mathrm{dB}$ |
| PSRR @ 1 MHz | $>2\,\mathrm{dB}$ |
| PSRR @ 10 MHz | $>2\,\mathrm{dB}$ |
| Output Referred Noise @ 10 Hz | $<-115\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output Referred Noise @ 1 kHz | $<-125\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output Referred Noise @ 10 kHz | $<-134\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output Referred Noise @ 100 kHz | $<-143\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output Referred Noise @ 1 MHz | $<-155\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |

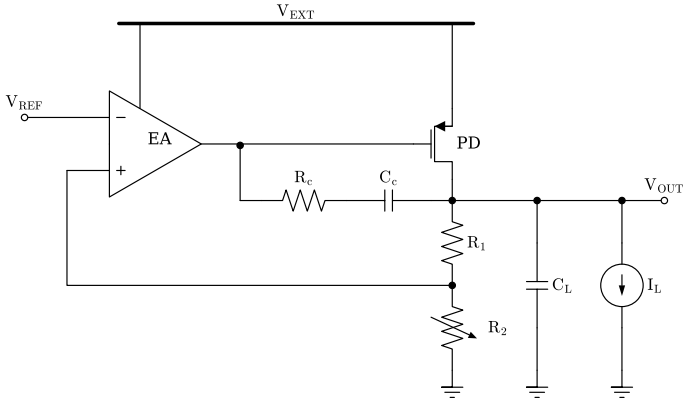



Fig. 1. General LDO Schematic

As shown in Figure 1, a PMOS LDO is made of an Error Amplifier (EA), a PMOS Pass Device (PD), two feedback resistors $R_1$ and $R_2$ for setting the output voltage and a compensation network consisting of a resistor $R_c$ and a capacitor $C_c$. In this work only one-stage EAs will be presented and analyzed.

### A. Output Voltage

Assuming that the EA has infinite DC gain (therefore $A_{EA,0} \to \infty$), LDO's gain is given by:

$$\frac{V_{OUT}}{V_{REF}} = \frac{A_{EA,0}}{1 + A_{EA,0}\beta} \xrightarrow{A_{EA,0}\to\infty} \frac{V_{OUT}}{V_{REF}} = \frac{1}{\beta} \tag{1}$$

where $\beta$ is the feedback factor set by $1 + \frac{R_1}{R_2}$. Therefore:

$$\frac{V_{OUT}}{V_{REF}} = \frac{R_1}{R_1 + R_2} \to \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1 \tag{2}$$

### B. Stability

A small signal model of the LDO schematic can be seen in Figure 2 where the EA's transconductance, output resistance and output load are represented by $g_{m,EA}$, $R_1$ and $C_1$ respectively. Similarly, $g_{m,p}$, $R_2$ and $C_2$ represent the same values for the PD. The LDO's transfer function has two poles (given that we have a one stage EA), $\omega_{p1}$ (dominant) and $\omega_{p2}$ (non-dominant) and a zero $\omega_z$, all of which can be approximated with:

$$\omega_{p1} = \frac{1}{g_{m,p} R_1 R_2 C_c} \tag{3}$$

$$\omega_{p2} = \frac{g_{m,p}}{C_1 + C_2} \tag{4}$$

$$\omega_z = \frac{-1}{C_c(1/g_{m,p} - R_c)} \tag{5}$$

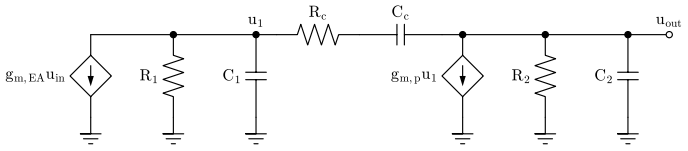



Fig. 2. Small Signal Model of LDO

Even though unity gain bandwidth is not a required specification, one could find the unity gain frequency of the LDO using:

$$UGF \cong \frac{\beta g_{m,EA}}{C_2} \tag{6}$$

### C. Power Supply Rejection Ratio (PSRR)

Power supply rejection refers to the amount of voltage ripple at the output of the LDO coming from the input voltage and therefore is of great importance. The PSR transfer function of a LDO regulator strongly depends on the type of the EA and whether the input transistors of the EA are NMOS or PMOS (Type-A or Type-B EA respectively). For a PMOS PD, Type-A EAs give the best PSRR results at low frequencies [2]. Therefore, all the LDOs presented in this work utilize Type-A EAs and their PSRR can be approximated by:

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,p} r_{ds,p}} \tag{7}$$

At higher frequencies, the compensation resistor can help in acheiving the PSRR requirement.

## REFERENCES

[1] T. Chan Carusone, D. A. Johns, and K. W. Martin, *Analog Integrated Circuit Design*, 2nd ed. John Wiley & Sons, Inc., 2012.

[2] J. Torres, M. El-Nozahi, A. Amer, S. Gopalraju, R. Abdullah, K. Entesari, and E. Sanchez-Sinencio, "Low Drop-Out Voltage Regulators: Capacitorless Architecture Comparison," *IEEE Circuits and Systems Magazine*, vol. 14, no. 2, pp. 6–26, 2014.

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